

# Continuity: $\epsilon$ - $\delta$ and Sequential Characterisations

Math Foundations — Node 11

## 1 Definitions

Let  $(X, d_X)$  and  $(Y, d_Y)$  be metric spaces. We routinely specialise to  $X = Y = \mathbb{R}$  with  $d(x, y) = |x - y|$ .

**Definition 1** (Continuity at a point). *A function  $f : X \rightarrow Y$  is continuous at  $c \in X$  if for every  $\epsilon > 0$  there exists  $\delta > 0$  such that*

$$d_X(x, c) < \delta \implies d_Y(f(x), f(c)) < \epsilon.$$

*$f$  is continuous on  $X$  if it is continuous at every point.*

**Definition 2** (Uniform continuity).  *$f$  is uniformly continuous on  $X$  if for every  $\epsilon > 0$  there exists  $\delta > 0$  such that for all  $x, y \in X$ ,*

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \epsilon.$$

*The single  $\delta$  must serve every pair of points; the quantifier order distinguishes this from pointwise continuity.*

**Definition 3** (Lipschitz continuity).  *$f$  is Lipschitz with constant  $L \geq 0$  if*

$$d_Y(f(x), f(y)) \leq L d_X(x, y) \quad \text{for all } x, y \in X.$$

The trivial chain of implications holds:

$$\text{Lipschitz} \implies \text{uniformly continuous} \implies \text{continuous}.$$

None of the converses hold:  $\sqrt{x}$  on  $[0, 1]$  is uniformly continuous but not Lipschitz;  $x^2$  on  $\mathbb{R}$  is continuous but not uniformly continuous.

## 2 Main theorem

**Theorem 4** ( $\epsilon$ - $\delta$  and sequential continuity are equivalent). *Let  $f : X \rightarrow Y$  be a function between metric spaces and let  $c \in X$ . The following are equivalent.*

- (A)  *$f$  is continuous at  $c$  in the sense of Definition ??.*
- (B) *For every sequence  $(x_n) \subseteq X$  with  $x_n \rightarrow c$ , we have  $f(x_n) \rightarrow f(c)$  in  $Y$ .*

*Proof.* **(A)  $\Rightarrow$  (B).** Assume  $f$  is  $\epsilon$ - $\delta$  continuous at  $c$  and let  $(x_n)$  be any sequence with  $x_n \rightarrow c$ . Fix  $\epsilon > 0$ . By (A), choose  $\delta > 0$  such that

$$d_X(x, c) < \delta \implies d_Y(f(x), f(c)) < \epsilon. \quad (\star)$$

Since  $x_n \rightarrow c$ , there exists  $N \in \mathbb{N}$  with  $d_X(x_n, c) < \delta$  for all  $n \geq N$ . For such  $n$ ,  $(\star)$  gives  $d_Y(f(x_n), f(c)) < \epsilon$ . Hence  $f(x_n) \rightarrow f(c)$ , establishing (B).

**(B)  $\Rightarrow$  (A).** We argue by contrapositive: suppose  $f$  is *not*  $\epsilon$ - $\delta$  continuous at  $c$ . Then there exists  $\epsilon_0 > 0$  such that for every  $\delta > 0$ , some  $x \in X$  satisfies  $d_X(x, c) < \delta$  yet  $d_Y(f(x), f(c)) \geq \epsilon_0$ .

Specialise to  $\delta = 1/n$  for each  $n \in \mathbb{N}$ : pick  $x_n \in X$  with

$$d_X(x_n, c) < \frac{1}{n} \quad \text{and} \quad d_Y(f(x_n), f(c)) \geq \epsilon_0.$$

The first inequality forces  $x_n \rightarrow c$ . The second forces  $f(x_n) \not\rightarrow f(c)$  (otherwise the distances would eventually be smaller than  $\epsilon_0$ ). This violates (B), establishing the contrapositive and hence (B)  $\Rightarrow$  (A).  $\square$

### 3 Consequences

**Lemma 5** (Composition). *If  $f : X \rightarrow Y$  is continuous at  $c$  and  $g : Y \rightarrow Z$  is continuous at  $f(c)$ , then  $g \circ f$  is continuous at  $c$ .*

*Proof.* By Theorem ??, continuity is preserved along sequences: if  $x_n \rightarrow c$  then  $f(x_n) \rightarrow f(c)$ , hence  $g(f(x_n)) \rightarrow g(f(c))$ . Apply Theorem ?? in the reverse direction.  $\square$

**Theorem 6** (Heine–Cantor, stated). *If  $K \subseteq \mathbb{R}^d$  is compact and  $f : K \rightarrow \mathbb{R}$  is continuous, then  $f$  is uniformly continuous on  $K$ .*

The proof (omitted) follows the contrapositive recipe of Theorem ??: failure of uniformity yields sequences  $(x_n), (y_n)$  with  $|x_n - y_n| \rightarrow 0$  yet  $|f(x_n) - f(y_n)| \geq \epsilon_0$ ; compactness extracts a convergent subsequence, contradicting pointwise continuity.

### 4 Example

**Example 7** ( $x^2$  at  $x = 2$ ). *Fix  $\epsilon > 0$ . For  $|x - 2| < 1$  we have  $|x + 2| < 5$ , so*

$$|x^2 - 4| = |x - 2| |x + 2| < 5 |x - 2|.$$

*Choose  $\delta = \min(1, \epsilon/5)$ . Then  $|x - 2| < \delta \Rightarrow |x^2 - 4| < \epsilon$ , verifying Definition ??.*